Kaushik Tummalapalli

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Summary

Senior Machine Learning Engineer with 2+ years of experience designing and deploying production systems and inference pipelines. Specialized in LLM-powered applications, ML platform engineering, and large-scale unstructured data processing. Strong background in Python, distributed systems, evaluation frameworks, and cost–performance optimization.

Technical Expertise

- **Languages:** Python, Java, SQL, C++
- **AI/ML & Generative AI:** Large Language Models (LLMs), NLP, PyTorch, Scikit-learn, HuggingFace, RAG, Experiment Tracking, Model Evaluation & Monitoring, Time Series Forecasting, Agentic Workflows, Recommendation Systems, MLflow
- **Data & Distributed Computing:** PySpark, Snowpark, Large-Scale Data Processing, Batch ML pipelines
- **Cloud, Infrastructure & MLOps:** Google Cloud Platform (Vertex AI, GKE, Cloud Functions), Azure, Docker, CI/CD, Git
- **Backend / APIs:** FastAPI, Flask, REST APIs, Microservices

Work Experience

CVS Health **May 2024 – Present**
Senior Machine Learning Engineer United States

- Designed and deployed **cloud-scale ML inference services in Python**, architecting **OCR + LLM multi-stage pipelines** processing **150K+ unstructured documents**, achieving **sub-second p95 latency**, **high availability (99.9% uptime)**.
- Spearheaded transition from **custom neural models to LLM-based extraction**, improving **entity coverage from 65% to 95%** and delivering ~**$350K in annual cost savings** through reduced model maintenance and retraining overhead.
- Built an **LLM-as-Judge evaluation framework**, achieving **90% recall** in detecting incorrect model outputs and enabling **automated explainability and quality monitoring**.
- Prototyped and productionized **automated evaluation and ML pipeline infrastructure**, architecting a multi-stage, multi-LLM document extraction platform with orchestration, evaluation, fallback strategies, monitoring, and cost–performance optimization, enabling **50% automated error detection**.
- **Engineered a 0→1 asynchronous OCR microservice (FastAPI + GKE)**, implementing **Kafka-driven async orchestration** and **multi-provider OCR routing**, enabling **10K+ requests/day** with stable production reliability.
- Partnered closely with **product managers, platform engineers, privacy, and domain stakeholders** to translate ambiguous requirements into reliable, user-centric AI solutions aligned with health and wellness use cases.

CVS Health **May 2023 – Aug 2023**
Machine Learning Engineer Intern United States

- Built and productionized **end-to-end forecasting pipelines** using **Snowpark and Snowflake**, processing **multi-year cloud cost data** across ingestion, transformation, feature engineering, training, and deployment.
- **Implemented and benchmarked forecasting models** (Prophet, ARIMA, SARIMA, XGBoost), achieving ~**20% MAPE** for cloud cost prediction and financial planning use cases.
- Developed **reproducible ML experimentation workflows** and integrated predictions into a **Streamlit analytics application**, improving cloud spend visibility and supporting data-driven decision-making.

Zemoso Technologies **Dec 2021 – May 2022**
Software Engineer Intern Hyderabad, India

- Built a **production-grade web application** using **React** and **JavaScript**, delivering core user workflows end-to-end and achieving **90%+ automated test coverage**, improving reliability and reducing regressions.
- Applied **software engineering best practices** including **Git-based version control, code reviews, CI workflows**, and **agile development** (sprint planning, reviews, iterative delivery), collaborating cross-functionally to reduce **time-to-market by 24%**.

Education

New York University **Sep 2022 – May 2024**
Master of Science, Computer Engineering New York, United States
Relevant Courses: Machine Learning, Big Data, Data Science (CDS), Software Engineering, Algorithms, Data Structures, Systems Design.

Projects

Cloud-to-Runtime Optimization of LLM Inference Pipelines

- Built and optimized an end-to-end LLM inference pipeline for headline generation using a **fine-tuned Llama-1B-scale model**, applying KV-caching, quantization, pruning, speculative decoding, and DeepSpeed parallelism, and benchmarking latency, throughput, memory, and ROUGE across cloud and **LiteRT-compatible** runtime deployments.

UdaPlay — AI Research Agent (RAG + Web Search Fallback + Memory)

- Built a **stateful LLM agent using RAG** over a persistent vector database (1K+ game entities) with automatic web-search fallback triggered by retrieval confidence, improving answer completeness by **40% across evaluated queries**; generated well-cited, structured responses and persisted newly discovered facts into long-term memory for reuse across sessions.